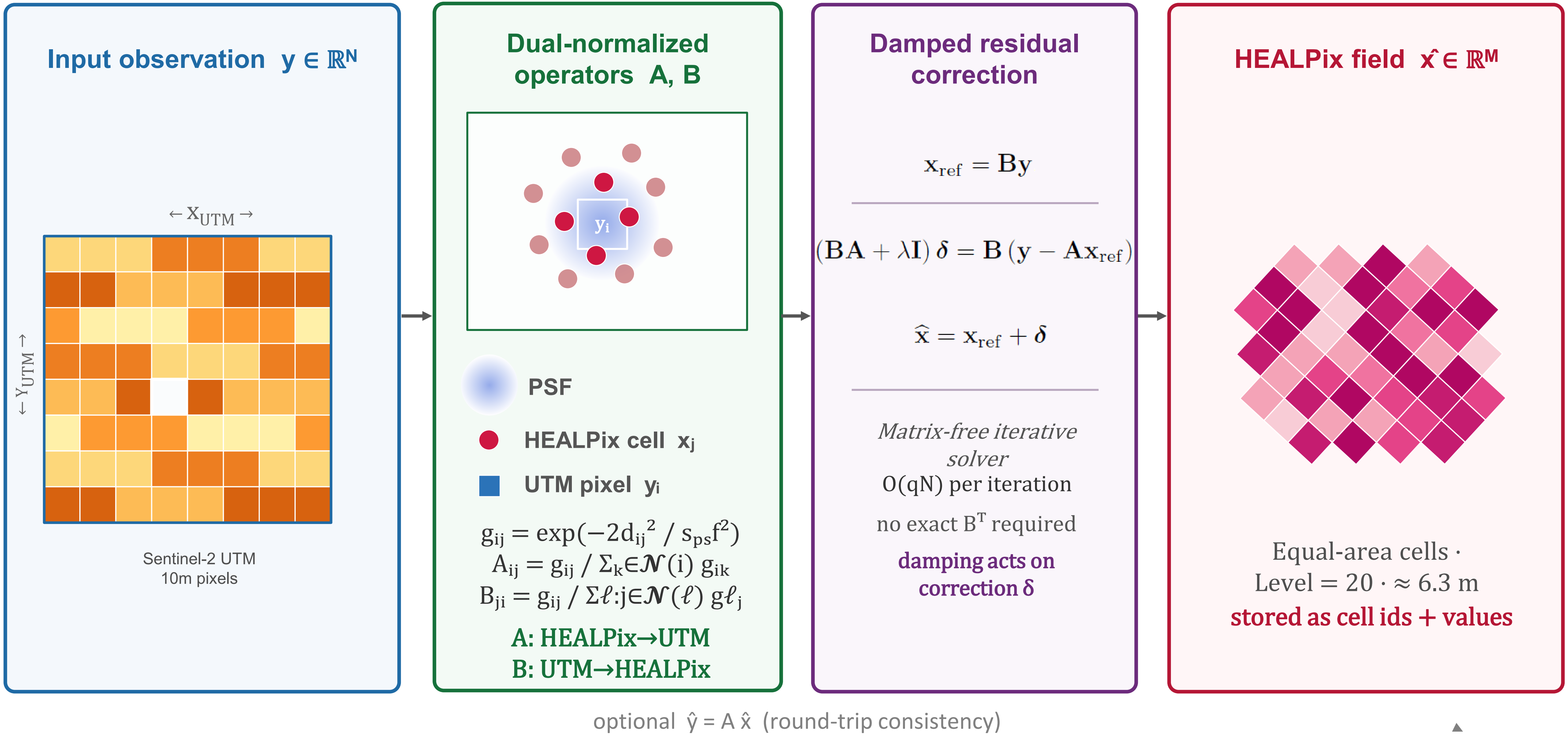
Input observation y ∈ ℝ^N
← x_UTM →
← y_UTM →
Sentinel-2 UTM
10m pixels
Dual-normalized operators A, B
y_i
PSF
HEALPix cell x_j
UTM pixel y_i
g_ij = exp(−2d_ij^2 / s_ps f^2)
A_ij = g_ij / Σ_k∈𝒩(i) g_ik
B_ji = g_ij / Σℓ:j∈𝒩(ℓ) gℓ_j
A: HEALPix→UTM
B: UTM→HEALPix
Damped residual correction
x_ref = By
(BA + λI) δ = B (y − Ax_ref)
x̂ = x_ref + δ
Matrix-free iterative solver
O(qN) per iteration
no exact B^T required
damping acts on correction δ
HEALPix field x̂ ∈ ℝ^M
Equal-area cells ·
Level = 20 · ≈ 6.3 m
stored as cell ids + values
optional ŷ = A x̂ (round-trip consistency)